

Jeffrey "Jack" Hannon Jr.

Linux / HPC Systems Administration · Automation · Diagnostics

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Summary

Linux-focused systems administrator working in a university HPC environment.

Day-to-day work includes Slurm-driven cluster operations, hardware triage, firmware/BIOS coordination with vendors, and automation with shell and Python. I build tools to reduce repetitive operational pain and make cluster behavior more visible.

Skills

Systems & Infrastructure: Linux (Debian, Rocky, Ubuntu), SLURM, Lustre, OFED, QEMU/KVM, Docker, BIOS/firmware coordination

Networking & Security: iptables, WireGuard, policy routing, DNS/DHCP (Pi-hole, dnsmasq), PKI/TLS (step-ca, ACME, mTLS)

Automation & Development: Python (curses, PTY, PySide6, pytest), Bash/Zsh, Ansible, Git

Platforms & Tooling: cloud-init, Forgejo/GitHub, ReFrame, LLM coding tools (Claude Code, Copilot)

Experience

Computer Specialist I — High-Performance Computing Center

Mississippi State University, Starkville, MS · 2022–Present

- Maintain and troubleshoot compute nodes across 4 Slurm clusters ranging from 96 to 1,800 nodes (including 16 GPU and 16 bigmem nodes): drain handling, node reboots, hardware diagnosis, and vendor RMA coordination, averaging 10–20 hardware failures per month.
- Coordinate firmware, BIOS, and OFED upgrades with Dell and internal teams, including IB adapter firmware managed through OpenManage Enterprise; run test jobs and document behavioral changes.
- Write and maintain diagnostic scripts (bash, Python) to surface node state, drain reasons, Lustre file system usage, and other health signals.
- Built and use bad, a personal terminal UI for daily node triage; its output is regularly shared with coworkers for analysis and tracing.
- Support roughly 5,000 researchers and staff with job failures, environment issues, and general HPC usage questions.

Research Computing Administrator (Freelance / Contract)

Remote · 2023–2024

- Maintained a bioinformatics software stack (~40 packages) supporting genomics and methylation sequencing workflows for a research group at Fox Chase Cancer Center.
 - Installed and built tools including Bowtie2, bwa-meth, MethylDackel, Picard, BUSCO, Augustus, HOMER, MaxQuant, and others using Spack where possible and manual source builds elsewhere.
 - Resolved dependency conflicts and build failures often working from incomplete or broken upstream documentation; sole technical contact for stack issues throughout.
 - Documented build procedures so environments could be rebuilt, and kept the stack current across upstream releases while maintaining legacy versions needed by existing workflows.
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Selected Projects

bad — HPC Administration Terminal UI

Tech: Python, curses, PTY, SLURM, NHC, iDRAC/RACADM, pytest

- Built a persistent interactive TUI replacing ad-hoc shell workflows for day-to-day cluster administration across large node sets
- Integrates SLURM, NHC, and Dell iDRAC RACADM into a single stateful session with live output capture, drainlist diff tracking, and cached sudo credentials
- ~8,500 lines across ~90 modules with 100+ pytest files including PTY simulation for integration-level testing

Home Network Lab — vplan & stepfam

Tech: Linux, WireGuard/NordVPN, iptables, Pi-hole, step-ca, Docker, Caddy, bash, Python

- Designed and maintain a multi-segment home network with a dual-NIC Debian Linux router providing LAN-wide VPN tunneling with selective per-device bypass via iptables mangle and policy routing tables
- Built vplan: a device-management toolkit generating Pi-hole DHCP reservations and DNS records idempotently from a single source-of-truth config, with full DNS postcheck validation
- Built stepfam: a two-part PKI toolkit managing a private ACME CA (step-ca in Docker) with inventory-driven cert issuance, SSH-based deployment, trust-chain distribution, and CA rotation — integrated with Caddy for automatic TLS on self-hosted services

netcon — iOS Network Coverage Mapper

Tech: Swift, SwiftUI, CoreLocation, CoreMotion, CoreData, MapKit, CoreGraphics

- Built a self-contained iOS app that passively measures Wi-Fi and cellular quality (latency, loss, throughput) as the user moves and renders results as a zoomable heatmap overlay
- Implemented a custom MapKit tile rasteriser with manual Web-Mercator projection, zoom-adaptive power-of-2 grid snapping, and confidence-based opacity blending
- Zero third-party dependencies; covers background location scheduling, Core Data persistence, and multi-format export (zip, CSV, GeoJSON)

myvm — Apple Silicon VM Manager

Tech: Python, PySide6, QEMU, QMP, HVF, cloud-init

- Desktop GUI for full VM lifecycle management on Apple Silicon using QEMU with HVF acceleration and macosvm for macOS guests
- Drives QEMU over QMP Unix socket for graceful ACPI shutdown; auto-manages SSH config blocks so VMs are immediately reachable by name after first boot
- Generates cloud-init NoCloud seed ISOs with Jinja-templated user-data for reproducible guest provisioning

Certifications/Awards

CompTIA A+

Awarded: 11/2025

Education

B.S. in Computer Science

Mississippi State University · 2021